SUBTOTAL Function in Excel

1. What is SUBTOTAL?

The SUBTOTAL function calculates a summary of a range, such as a sum, average, count, maximum, or minimum, and lets you control how hidden and filtered rows are treated. It solves the problem that SUM and AVERAGE still count rows you have filtered out. A Data Analyst uses it to get totals that match exactly what is visible on screen. Excel's Data > Subtotal command and a Table's Total Row use it behind the scenes.

2. Syntax

`=SUBTOTAL(function_num, ref1, [ref2], ...)`

Argument	Required?	What it accepts
`function_num`	Required	A number that chooses the calculation: 1 to 11, or 101 to 111 (see the table below)
`ref1`	Required	A range or cell reference to calculate (for example `B2:B5`)
`ref2, ...`	Optional	More ranges or references. Excel accepts up to 254 of them.

Calculation	SUM	AVERAGE	COUNT	COUNTA	MAX	MIN
function_num (includes manually hidden rows)	9	1	2	3	4	5
function_num (ignores manually hidden rows)	109	101	102	103	104	105

Most often you will use `=SUBTOTAL(9, B2:B5)` for a total, or `=SUBTOTAL(109, B2:B5)` to also skip rows you hid by hand.

3. Simple Example

	A	B
1	Student	Marks
2	Rahim	70
3	Karim	80
4	Hasan	60
5	Nabil	90
6	**Subtotal**	**300**

Formula: `=SUBTOTAL(9, B2:B5)`
Excel adds 70 + 80 + 60 + 90 and returns **Result: 300**. With no hidden rows, this matches SUM.

4. Another Example

	A	B
1	Product	Monthly Sales
2	Laptop	12000
3	Mouse	3500
5	Monitor	8000
6	Headset	6000
7	**Subtotal**	**29500**

Formula: `=SUBTOTAL(9, B2:B6)`
A filter hides row 4 (Keyboard, 4500). SUBTOTAL skips it: 12000 + 3500 + 8000 + 6000 = 29500. **Result: 29500**. A plain SUM would still give 34000.

5. When to Use It

- Total only the visible rows of a filtered sales list
- Count how many rows a filter is currently showing
- Find the average, maximum, or minimum of the filtered rows
- Add group subtotals to a list without double counting them in the grand total
- Build a report where you hide rows by hand and the total must update

6. Important Things to Remember

- SUBTOTAL applies the function you choose (SUM, AVERAGE, COUNT and so on) only to the rows that count.
- Codes 1 to 11 include rows you hid by hand. Codes 101 to 111 ignore them. Both always ignore rows hidden by a filter.
- Any other SUBTOTAL formulas inside the range are ignored, so a grand total never double counts.
- Hidden columns are not ignored by either set of codes. SUBTOTAL is built for rows of data in a column.
- If a referenced cell contains an error such as #N/A or #DIV/0!, the SUBTOTAL result shows that error.
- A function_num that is not valid (for example 12) or a 3-D reference returns `#VALUE!`.

7. Common Mistakes

Mistake: Using SUM on a filtered list. SUM still adds the rows the filter has hidden. **Fix:** Use SUBTOTAL so the total matches the visible rows.

Mistake: Mixing up 9 and 109. Code 9 still counts rows you hid by hand. **Fix:** Use 109 (or 101 to 111 for other functions) when hidden rows must not count.

Mistake: Expecting hidden columns to be ignored. Neither set of codes skips them. **Fix:** Filter rows instead, or point the formula only at the columns you want.

Mistake: Including the Subtotal cell itself, or an old Total row, in the range. **Fix:** Select only the data rows. Including the formula cell causes a circular reference.

8. Data Analyst Tip

Always compare the SUBTOTAL with the data it came from. Next to your total, calculate `=SUBTOTAL(103, A2:A6)` to see how many rows are visible, and compare it with `=COUNTA(A2:A6)` for all rows. If the visible count is lower, some rows are hidden, so your total covers only part of the dataset.